Art and Illusion

Thursday, 9 April 2026

ART IS ILLUSION. A painting or drawing is a flat surface covered with marks, yet those marks can be seen as bodies, faces, reflections, distance, glass, metal, or skin. The painted or drawn mark is never the thing itself. Art works because it prompts the viewer to experience marks as if they were real objects.

A painter of realism therefore does not simply copy what is seen. They construct an image that guide perception. Linear perspective, tonal modeling, reflected light, shadow, and optical detail are not ends in themselves. They are methods for causing the mind to assemble a convincing visual experience from marks on a plane.

Any painting or drawing can only partly succeed as an illusion. A flat image cannot reproduce the full richness of embodied sight. It cannot literally give us solid bodies. What it can do is evoke, in the mind of the viewer, an experience analogous to seeing real things.

This connection between art and illusion was recognized long ago. Pliny understood that seeing is not accomplished by the eye alone. The eye receives light, but the mind is the true instrument of vision. What we see is not merely what falls upon the retina. It is what the brain makes from the incoming pattern of light.

> the mind is the real instrument of sight and observation, the eyes act as a sort of vessel receiving and transmitting the visible portion of the consciousness.

EVERY DRAWING IS ALREADY AN ILLUSION. Real objects are not bordered by actual black lines. What we call an outline is a construction of the visual system. The eye registers differences in light reflected from adjacent regions of the world, and the brain converts those differences into perceived boundaries. A line placed at the edge of a form does not imitate its physical structure, but the way objects are segmented by vision.

Thus, drawing begins naturally with outline. To draw the contour of a face, a hand, or an eye leverages a basic operation of visual perception. The line is persuasive because the brain is disposed to treat boundaries as evidence of objects. The human brain transforms patterns of illumination – what happens when light is projected into the eye and onto the retina – into what we see.

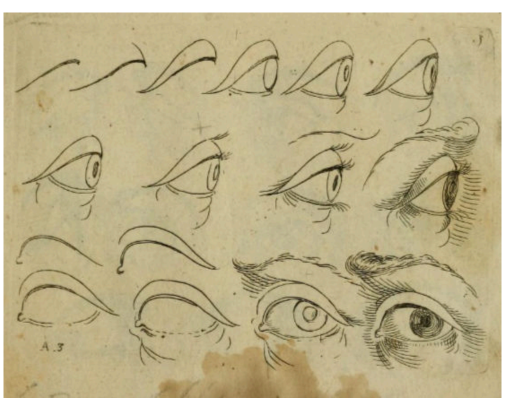


Figure 191: Odoardo Fialetti's drawing manual reduces the eye to a sequence of contours. The image works because the visual system readily accepts boundary lines as sufficient evidence for form.

We see objects above all by seeing edges. A boundary in the world often appears where there is a sudden change in light intensity between one region and the next. Within a single object, light may vary gradually across a surface. At the object's perimeter, by contrast, the change is often abrupt. The visual system treats such discontinuities as signs of separate things.

This is why black-and-white photography works so well. A monochrome photograph removes color altogether, yet the world remains legible. Trees, sky, water, stone, shadow, and reflection are still recognizable because objects are not primarily recognized by color. Objects are defined by contrast, contour, and the organization of regions by light.

A photograph in black and white demonstrates a fundamental principle. Vision does not require a complete copy of reality in order to recognize reality. A sufficiently organized visual pattern is enough.

Figure 192: Carleton Watkins, Yosemite. Even without color, the image remains fully intelligible because the visual system identifies objects through contrast, edge, and spatial organization.

THE PHYSIOLOGY OF VISION helps explain why line and contrast are so powerful. The retina is not a neutral recording surface like photographic film. It is a sheet of neural tissue that begins the work of interpretation before visual signals ever reach the brain.

Santiago Ramón y Cajal, the great anatomist of the nervous system, was the first to map the cellular organization of the retina in detail. His drawings show extensive lateral connections among neighboring regions. We now understand that such circuitry contributes to edge detection. When one area of the retina is strongly stimulated by light, nearby regions are suppressed. The net effect enhances local contrast at boundaries.

This process is often described as center-surround inhibition. It allows the visual system to register not how much light is present, but where light changes sharply across space. The retina emphasizes places where one object ends and another begins.

This is why a drawn line can be compelling. From the beginning, the visual system is converting contrast into contour. Once an boundary is established, the mind supplies its form.

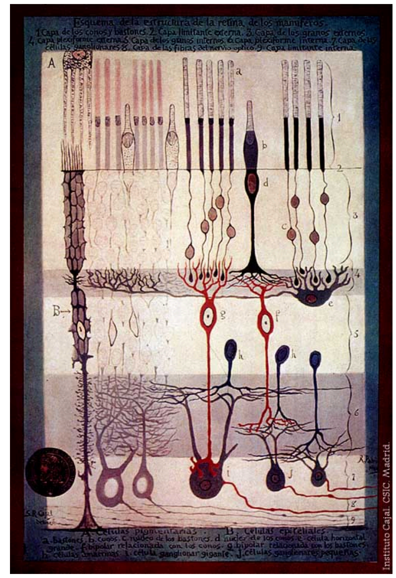


Figure 193: *Structure of the Mammalian Retina* c.1900 By Santiago Ramon y Cajal. a. Rods. b. Cones. c. Rod nucleus. d. Cone Nucleus. e. Large horizontal cell f. Cone-associated bipolar cell. g. Rod-associated bipolar cell. h. Amacrine cells. i. Giant ganglion cell. j. Small ganglion cells.

Figure 194: An edge filter in computer vision that is based on lateral inhibition like the retina. From Wikipedia.

The visual system does not merely detect boundaries. It also exaggerates them. One consequence is the phenomenon known as Mach bands. When adjacent regions of uniform brightness meet at a sharp boundary, the darker side appears slightly darker near the edge, while the lighter side appears slightly lighter. These bands are not present in the stimulus itself. They are generated by the operations of the visual system.

Mach bands matter because they show that seeing is not passive but involves active computation. The visual system heightens contrast at boundaries in order to clarify structure, and in doing so it also produces visual illusions.

Figure 195: Mach bands. The apparent darkening and lightening near each boundary is generated by visual processing, not by actual bands painted into the image.

Artists can exploit this fact. Jesse Stillwell, for example, uses sharp juxtapositions of flat color in ways that cause the eye to perceive gradients that are not literally painted. The painting works because the visual system cannot help but follow its own neural rules.

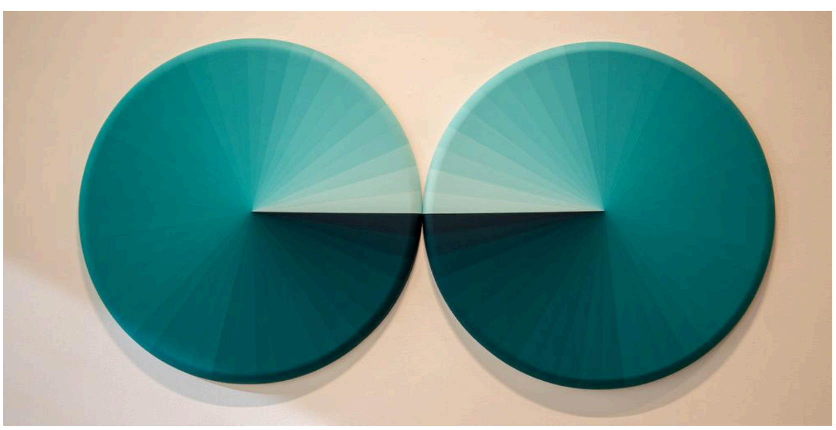


Figure 196: *Smack Dab in the Middle of Spectrum* by Jesse Stillwell, 2022. Acrylic on shaped canvas. 30"x60"

What these examples show is that visual experience is constructed at many levels. The mind draws boundaries where there are changes in light. It exaggerates contrast at those boundaries. It organizes incomplete information into coherent objects. A painting succeeds not when it copies everything, but when it gives the visual system enough information to complete the world for itself.

Johannes Vermeer's *Girl with a Pearl Earring* offers another kind of illusion. The painting is often described as a *tronie*, not a conventional portrait of a known individual but a study of expression, costume, and type. Whether the girl corresponds to any real sitter matters less than the fact that Vermeer makes us experience a convincing human presence through remarkably selective means.

The image appears full and immediate, yet Vermeer does not describe everything equally. He withholds as much as he gives. He paints only those cues required to give the impression of a face turning in light, of moist eyes, lips, skin, and a luminous pearl suspended at the ear.

Realism in painting does not depend on exhaustive transcription. It depends on the precise placement of cues to which the visual system is especially responsive.

Figure 197: Johannes Vermeer, *Girl with a Pearl Earring*, c. 1665, Mauritshuis, The Hague.

High-resolution imaging makes Vermeer's economy of means clear. The pearl is not rendered by minute descriptive detail. Vermeer suggests it with only a few strategically placed strokes: a bright highlight, a softer reflection below, and the surrounding tonal context that allows the eye to read the pearl as a smooth, rounded, and lustrous object.

The mind completes the pearl. It is seen not because every optical fact has been copied, but because enough have been supplied that the mind can complete recognition.

Figure 198: Detail of the pearl in Vermeer's *Girl with a Pearl Earring*. The pearl is not described point by point; a few strategically placed highlights and tonal relations are enough for the mind to recognize a smooth, lustrous sphere.

The same is true of the nose. Look carefully at the bridge of the nose and the transition into the cheek. Vermeer does not paint a hard contour. The form is inferred from adjacent shadow and light. The viewer sees a nose where painted information is surprisingly absent.

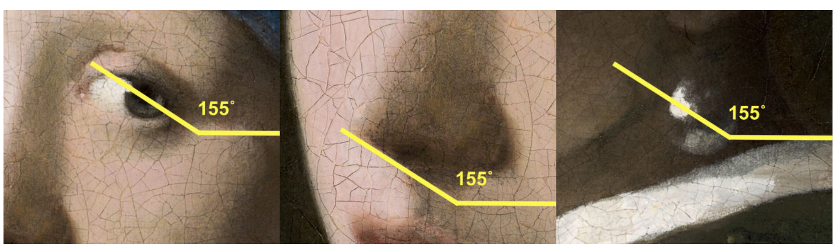
Technical analysis confirms how deliberately these effects were built. Imaging reveals underlayers, fine contours, and revisions that helped Vermeer control the final appearance of surfaces and edges. But the crucial lesson is simple: the painting works because the viewer does not passively receive images of objects. The viewer completes perception by active mechanisms.

Figure 199: Detail of the bridge of the nose. Vermeer avoids a hard contour, allowing adjacent light and shadow to imply projecting form. The nose is perceived through inference as much as through explicit drawing.

Digital imaging and analysis reveals how Vermeer made his painting: layer by layer, correction by correction, with extraordinary patience. He did not simply place a final image onto the canvas in one act of transcription. He built the illusion slowly. Vermeer laid down multiple underlayers that varied in tone, allowed them to dry, and then returned to them with thinner or thicker upper layers in different regions of the painting. The final effects of light and shadow were accumulated through a sustained process.

High-resolution imaging also reveals preparatory and corrective work beneath the finished surface. Fine black outlines helped establish contours in advance, and pentimenti show that Vermeer adjusted the image as he worked, relocating features such as the iris and the ear. These are not incidental corrections. They show a painter repeatedly testing what the eye would see, then refining the image until the desired perceptual effect was achieved.

Geometric analysis makes the same point at another level. The reflections in the eyes and pearl, and the shadow cast by the nose, all align with the same incoming direction of light. Such consistency is not accidental. It shows Vermeer checking the internal logic of illumination across the whole picture. The painting's immediacy is itself deceptive, containing hidden evidence of obsessive construction.



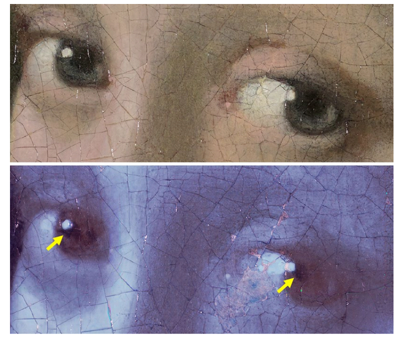


Figure 200: Evidence for pentimenti in the Girl's eyes. a Visible light photograph. b MS-IRR false colour detail. Dark marks indicate possible earlier iris locations (yellow arrows)

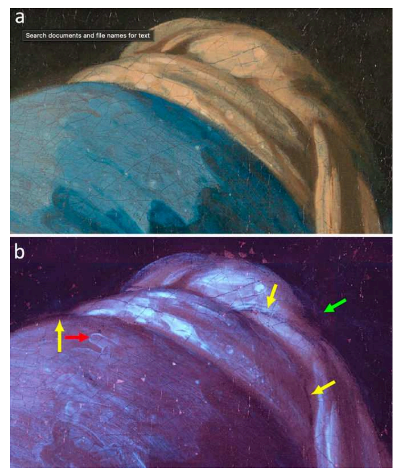


Figure 201: Evidence for contours and fine black outlines in the top of headscarf. a Visible light photograph. b MS-IRR false colour detail. Wavy brushstrokes at the surface (red arrow), fine black outlines applied in a preparatory phase (yellow arrows), back of the headscarf applied on top of an infrared-absorbing layer (green arrow).

Before reading further, look carefully at the image below. In this orientation it appears to be a conventional still life: an arrangement of edible things laid out for inspection.

Do not turn the book yet. Let the image first remain what it appears to be.

Then turn the page upside down.

The still life has become a face! Nothing has been added to the picture. No new contour has appeared, no new color has been painted, no new object has been added. Only the orientation has changed, and with it the brain's interpretation of the same forms.

Arcimboldo makes visible a deep fact about vision. The eye does not merely deliver a list of local features. The visual system searches constantly for coherent wholes, and among the most privileged of those wholes is the human face.

Figure 202: Giuseppe Arcimboldo, *The Vegetable Gardener* (*L'Ortolano*), ca. 1587–1590, Museo Civico Ala Ponzone, Cremona. In this orientation the same painted forms are reorganized as a face. Without adding a single new mark, the still life becomes a portrait, demonstrating that perception does not merely register parts but actively organizes them into a coherent whole.

Arcimboldo's paintings allow two readings at once. We can inspect them part by part, recovering individual objects. But we can also see them holistically, as a face that appears all at once. The whole becomes more than the sum of its parts.

Pictorial illusion is not a matter of imitating light, texture, or perspective. It is a matter of activating the mind's cognitive processes. A demonstration of the principle that Arcimboldo leverages is the so-called Thatcher illusion. In this illusion, a face is manipulated by turning the eyes and mouth upside down within the head. When the entire image is itself inverted, the distortions are strangely difficult to detect. But when the head is returned to its normal upright position, the face suddenly appears grotesque.

Figure 203: The Thatcher illusion. When the whole face is inverted, grotesque inversions of the eyes and mouth are surprisingly difficult to detect. When the head is returned to its normal upright orientation, the distortions become immediately apparent. Upright face perception is therefore unusually sensitive to the configuration of features.

Bibliography

- Ernst Gombrich. *Art and Illusion: A Study in the Psychology of Pictorial Representation*. Princeton University Press, 2000. ISBN 0691070008
- David G Stork. *Pixels & paintings : foundations of computer-assisted connoisseurship*. Wiley, Hoboken, New Jersey, 2024. ISBN 9780470229446
- Abbie Vandivere, Annelies van Loon, Kathryn A. Dooley, Ralph Haswell, Robert G. Erdmann, Emilien Leonhardt, and John K. Delaney. Revealing the painterly technique beneath the surface of Vermeer's Girl with a Pearl Earring using macro- and microscale imaging. *Heritage Science*, 7:64, 2019

On Photography

Thursday, 16 April 2026, 12:45 - 2:15 PM EST.

Figure 204: Daguerrotype of Henry Fox Talbot, 1844, by Antoine Claudet

.

Photography emerged not as a break from painting and drawing, but as the next chapter in a longer history of making appearances visible. The camera obscura had already performed an essential part of that work by projecting the visible world onto a plane. Talbot and Daguerre, the inventors of photography, added permanence. They did not invent projection itself; they invented a way for projected light to leave a durable trace.

William Henry Fox Talbot (1800–1877) wrote that he invented photography out of frustration with the camera lucida. In 1833, while trying to draw scenes from a vacation in Italy, he found that the device did not spare him the difficulty of draftsmanship. The resulting sketches were such that the *"faithless pencil had only left traces on the paper melancholy to behold."* He concluded that the camera lucida worked only for someone who could already draw.

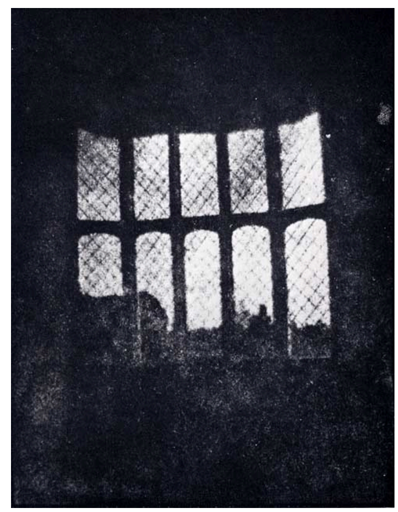


Figure 205: *Latticed window at Lacock Abbey* by Henry Fox Talbot, 1835.

But Talbot was also a scientist, and that mattered for the history of representation. In 1831 he had been elected to the Royal Society for his work in mathematics. He also knew chemistry and understood how light interacted with silver salts. When a solution of silver nitrate ($AgNO_3$) is mixed with sodium chloride ($NaCl$) and brushed onto paper, it forms a layer of light-sensitive silver chloride ($AgCl$). Light reduces $AgCl$ to metallic silver, darkening the paper where it is illuminated and creating a 'negative' image.

Talbot prepared paper with salt solutions in darkness. He exposed the paper to focused images and then 'fixed' the result. To fix an image is to stabilize it so that further exposure to light does not continue to darken the paper. Finally, Talbot reversed the image—turning negative to positive—by exposing a second sheet of paper through the first. His photograph of a latticed window, whose very name means 'light-writing' from Greek roots, became the first positive image made from a negative.

LOUIS DAGUERRE (1787–1851) found a different way to make photoreactive silver. He treated silver-plated copper plates with fumes that made their surfaces light-sensitive. After exposure, the image was developed with mercury vapor and then 'fixed' with chemicals. The resulting *daguerreotype* was a fragile image—a mere layer of atoms on a mirrored surface. When Daguerre announced his invention, Talbot also made his own work public.

Figure 206: Daguerreotype camera built by La Maison Susse Frères in 1839, with a lens by Charles Chevalier. From Wikipedia

THE CAMERA OBSCURA was the original housing for the methods of Talbot and Daguerre. A simple lens and darkened box could already project a focused image onto a surface, solving optically the same problem that Renaissance perspective had solved geometrically: how to translate depth into a flat image. The decisive new step in photography was not projection alone, but fixation. What painters had long done by hand—constructing a convincing likeness of the visible world on a plane—the camera now did with light itself. Modern cameras are still called 'cameras' even though they are no longer the 'dark rooms' of the Latin translation.

Figure 207: Louis-Jacques-Mandé Daguerre, *The Artist's Studio / Still Life with Plaster Casts*, 1837. One of the earliest surviving daguerreotypes, this image shows that photography began not only as a scientific process but also as an artistic one. Its carefully staged objects, dramatic light, and tactile surfaces place the new medium in direct conversation with the tradition of painted still life.

Talbot was inspired by the camera obscura before he invented the photographic camera. The optical principle was ancient. Aristotle and Leonardo da Vinci both knew that when rays of light pass through a pinhole, they project an image onto a distant surface. Talbot's excitement clarifies the larger logic of this course: perspective had taught artists how to construct a believable world on a flat surface, while the camera obscura seemed to offer that world already given, as if nature itself were drawing. Talbot described the first lightbulb of his idea:

> This led me to reflect on the inimitable beauty of the pictures of nature's painting which the glass lens of the Camera throws upon the paper in its focus—fairy pictures, creations of a moment, and destined as rapidly to fade away.
>
> It was during these thoughts that the idea occurred to me... how charming it would be if it were possible to cause these natural images to imprint themselves durably and remain fixed upon the paper!

Photography begins in this dream that realism might become automatic.

Abe Morell captures that dream in a modern photograph.

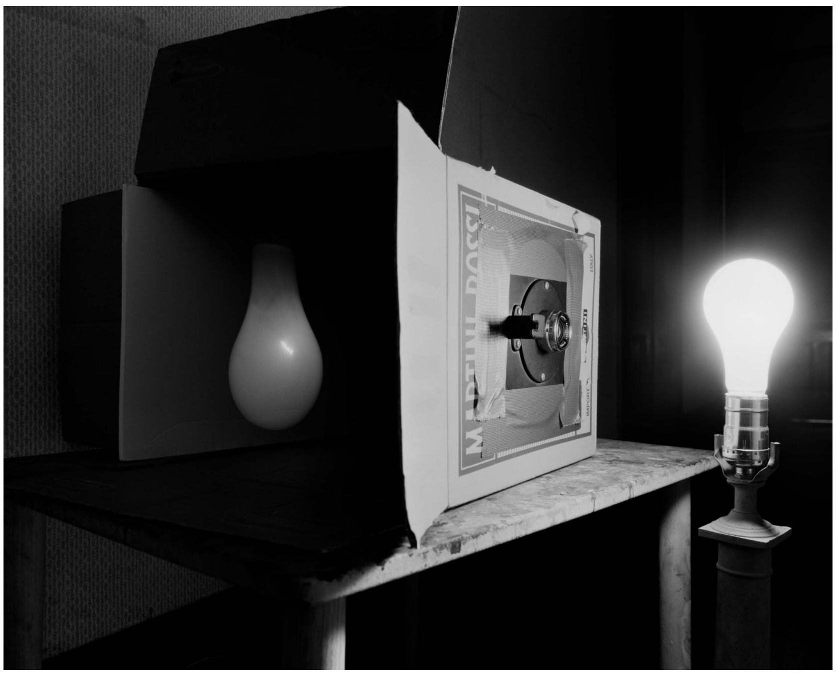


Figure 208: *Light Bulb*, Abe Morell, 1991. Archival pigment print. By making light itself seem to generate the image, Morell returns photography to its origins in the camera obscura and to the larger problem of how appearances become pictures.

Photography was invented, or so Talbot hoped, to subtract human fallibility from representation. Where perspective manuals had disciplined the hand, the camera promised to bypass it. Light itself would perform the labor of description. The early methods of Talbot and Daguerre were soon replaced by more practical chemistry and devices, but their larger ambition remained: a new kind of realism in which the visible world might seem to register itself. The first photographers pursued that ambition above all in nineteenth-century landscape and portraiture.

Carleton Watkins (1829–1916) was an early American photographer who made monumental views of the American West. He produced 'mammoth-plate' photographs of Yosemite, positive prints on a heroic scale that matched the grandeur of their subject. In Watkins, Talbot's dream of mechanical realism became sublime: the camera did not merely record the landscape, but gave it monumental presence. Photographic realism here approached the scale and ambition of history painting, even as it claimed to arise directly from nature itself.

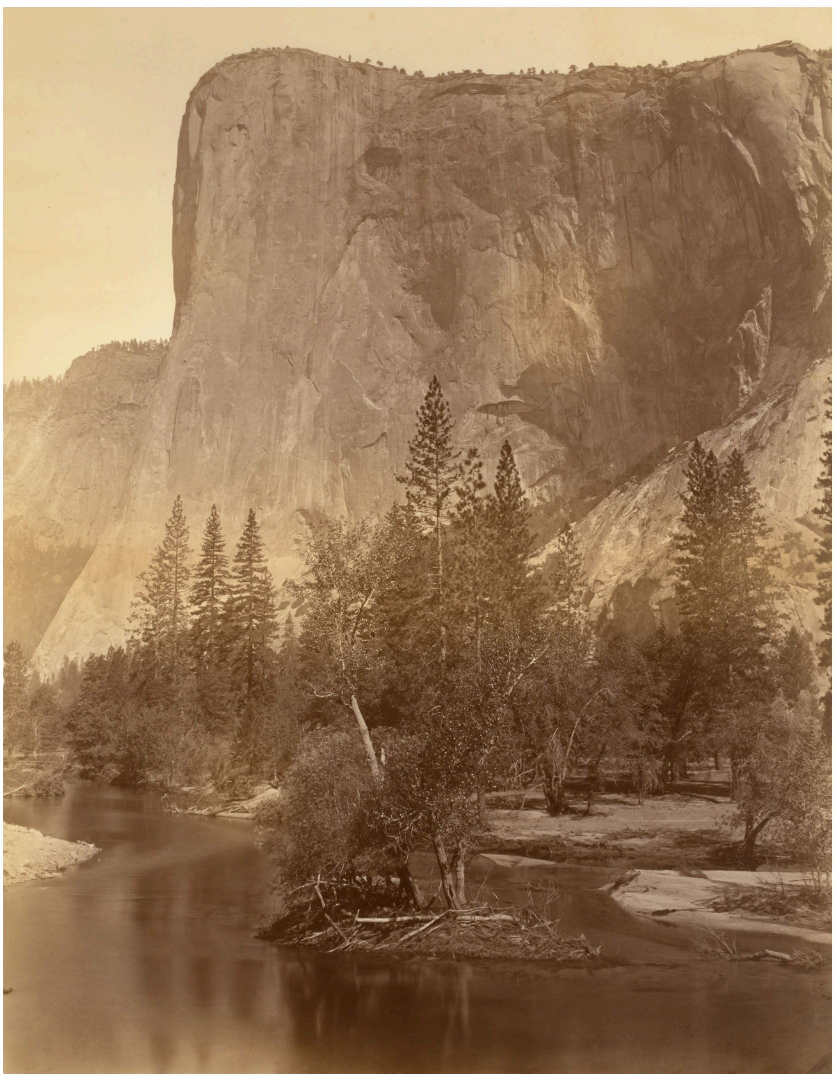


Figure 209: *El Capitan*, Carleton Watkins, 1860s. Mammoth-plate albumen silver print. Watkins used the descriptive power of photography to make the American landscape appear at once factual and sublime. Smithsonian American Art Museum

Julia Margaret Cameron (1815–1879) approached photography as though the new medium could inherit the highest ambitions of painting rather than define itself against them. Her camera did not turn away from older art; it reached back toward it. She had long studied Raphael, Giotto, and Michelangelo through prints circulating in nineteenth-century England. At the age of forty-eight, she took up photography at her home on the Isle of Wight, enlisting models from her family, servants, and neighbors. She wrote:

> My aspirations are to ennoble Photography and to secure for it the character and uses of High Art by combining the real and Ideal and sacrificing nothing of the Truth by all possible devotion to Poetry and beauty.

Accordingly, she chose idealized subjects, often staging her photographs as if they were paintings or poetic illustrations.

In Cameron's hands, the camera became less an instrument of mere recording than a device for reviving the ambitions of painting.

Figure 210: *And Enid Sang*, Julia Margaret Cameron, 1875. Albumen silver print. For Cameron, photography was not a wholly separate art, but a new vehicle for the older ambitions of painting: idealization, poetry, and the elevation of the human figure. Getty

Photography had new possibilities as devices improved and chemistry advanced. The first photographic exposures took hours, limiting pictures largely to landscape and still life. When exposures fell to minutes, photography expanded into posed portraiture. With hand-held cameras and sub-second exposures, photography took to the streets.

This shift had an earlier analogue in seventeenth-century Dutch painting. Long before hand-held cameras, Dutch artists had already turned from the more static and idealized modes of much Renaissance painting toward scenes of ordinary life, rendered with a new attentiveness to light, surface, and the impression of a moment. In some cases—most famously Vermeer—the camera obscura may have contributed to this heightened optical awareness, though it did not determine the paintings by itself. What hand-held photography later accomplished mechanically through fast exposure, Dutch painters had earlier pursued by artistic means: the appearance of immediacy, as if life had been caught in passing.

Alfred Stieglitz (1864–1946) looked for art in urban life. In 1892, he bought his first hand-held camera, a Folmer and Schwing 4x5 plate-film camera. It allowed him to capture the fleeting moment rather than the posed one. In 1893, he photographed steaming horses in front of the Old Post Office in Manhattan.

> Naturally there was snow on the ground. A driver in a rubber coat was watering his steaming horses. There seemed to be something related to my deepest feeling in what I saw, and I decided to photograph what was within me.

Figure 211: *The Terminal*, Alfred Stieglitz, 1893. Gelatin silver print. What Dutch painters had once achieved by making still images feel momentary, Stieglitz could now pursue with a camera quick enough to seize the fleeting life of the modern city. Art Institute of Chicago

PAINTING AND DRAWING changed in subject matter across centuries. Much early Renaissance painting in Italy was made for churches and chapels, and therefore often took idealized sacred subjects as its theme. By contrast, many Dutch painters of the seventeenth century turned toward domestic interiors, still life, landscape, and scenes of everyday life made for private homes. The shift granted a new dignity to the ordinary.

Photography followed a similar arc. The first generation of photographers often pursued elevated or idealized visions. Their successors increasingly found artistic meaning in the everyday.

EDWARD STEICHEN photographed a milk bottle on a tenement fire escape in New York in 1915. From his earlier painterly work, he shifted purpose:

> When I first became interested in photography, I thought it was the whole cheese. My idea was to have it recognized as one of the fine arts. Today I don't give a hoot in hell about that. The mission of photography is to explain man to man and each man to himself.

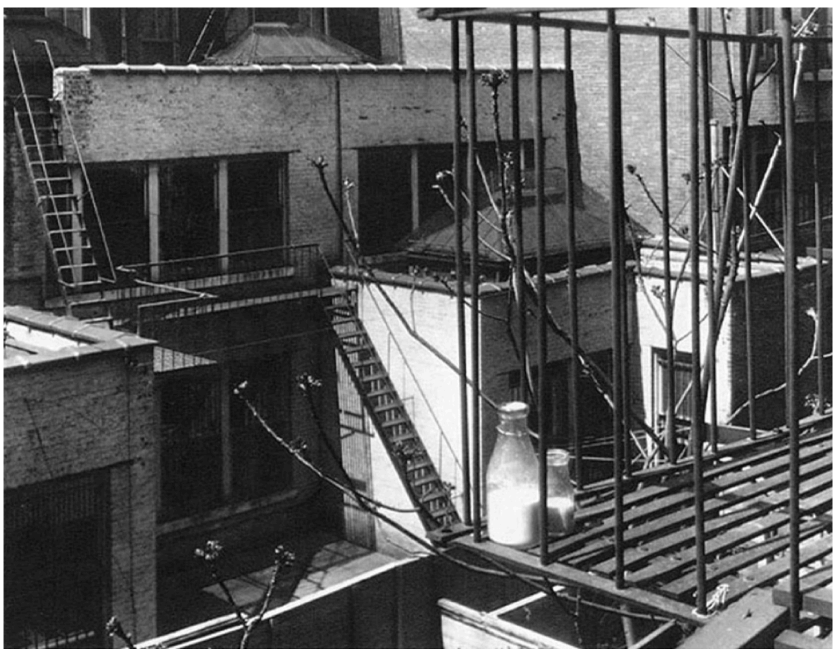


Figure 212: *Milk Bottles on a Fire Escape*, Edward Steichen, 1915. Steichen's photograph grants the ordinary world a new dignity, showing how photography, like Dutch painting before it, could make everyday life newly visible as art.

Diane Arbus (1923–1971) lived and died in New York City. Her photographs confront the human condition without consolation or idealization. She connected with people who had rarely been seen by photographers before, and who nevertheless allowed themselves to be seen by her.

Most of her portraits are frontal. Her subjects look straight into the camera, looking at Arbus as she looks at them. Because Arbus is looking through the camera, her subjects also seem to look at us as we look at them. Her photographs make the viewer newly self-conscious, turning seeing into a kind of voyeurism.

Arbus used inexpensive commercial film that allowed many exposures on a single reel. She could therefore photograph the same scene repeatedly and later choose among many possible images. Her contact sheet of *Child with a Toy Hand Grenade* shows that she selected the boy's most unsettling pose.

Photography does not erase the human from the making of a photograph. Beyond deciding what to photograph, the photographer decides how to frame, enlarge, crop, and print the image, which photographs become 'art,' and how they will be displayed.

Figure 213: Lady on a busy, NYC, by Diane Arbus, 1957. The Metropolitan Musem

Figure 214: *Contact sheet for Child with a Toy Hand Grenade in Central Park, N.Y.C.* Arbus's working sheet reveals that the final image was chosen from many exposures, making visible the photographer's act of selection behind the apparent immediacy of the photograph.

Figure 215: *Child with a Toy Hand Grenade in Central Park, N.Y.C.*, Diane Arbus, 1962. Arbus's famous image feels immediate and unguarded, yet her contact sheet shows that even documentary photography depends on selection: the photograph is made not only when the shutter clicks, but also when the photographer chooses which moment will stand for the scene. The Metropolitan Museum of Art

As PHOTOGRAPHY CHANGED from the nineteenth to the twentieth century, painting and drawing changed with it, partly in response to photography's new claim on realism. Once the photograph seemed to secure realism mechanically, painting was newly free—or newly compelled—to do what photography could not. The Impressionists pursued sensation rather than exact contour. The Surrealists imagined scenes that had never been seen. The Modernists, in the most radical cases, imagined painting without scenes at all.

PHOTOGRAPHY SEEMED TO SOLVE THE AGE-OLD PROBLEM of linear perspective by optical means rather than manual construction. But photography did not abolish the old problem of perspective so much as inherit it in another form, because every photograph still translates a three-dimensional world into a two-dimensional image. In the nineteenth and twentieth centuries, scientists were redefining space and dimension at the same moment that artists were redefining the aims of representation. A square, for example, can be understood as the two-dimensional shadow of a three-dimensional cube. The cube, in turn, may be imagined as the three-dimensional shadow of a four-dimensional tesseract. Even higher dimensions can be imagined, but they too must be flattened onto two-dimensional images in order to be seen as pictures. In *Corpus Hypercubus*, the Surrealist Salvador Dalí (1904–1989) depicts Christ crucified on a tesseract.

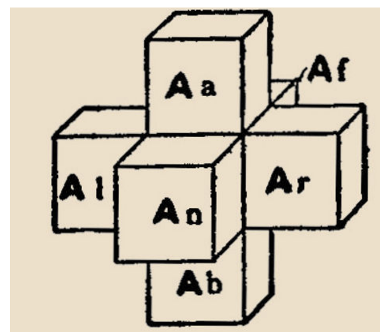


Figure 216: *Hypercube*, Charles Hinton, 1912. A diagram of the tesseract, or four-dimensional analogue of the cube, flattened into a form that can be seen on a page.

Figure 217: *Crucifixion (Corpus Hypercubus)*, Salvador Dalí, 1954. Dalí uses the unfolded tesseract to suggest a space beyond ordinary perception, showing how even visions of higher dimension must still be rendered in flat pictorial form. The Metropolitan Museum of Art.

Susan Sontag (1933–2004) wrote *On Photography*, her seminal meditation on the camera's effect on modern visual culture. We live in a world where nearly everything that can be photographed seems already to have been photographed. Yet the visual reach of photography and art is not infinite. We see what we are shown.

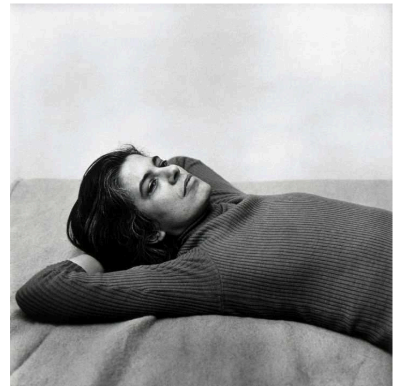


Figure 218: *Susan Sontag*, Peter Hujar, 1975. Sontag became one of photography's most incisive interpreters, asking not only what photographs show, but how they shape what a culture is able to see. The Metropolitan Museum of Art

The Allegory of the Cave in Plato's *Republic* is an older metaphor for the limits of visual experience. Socrates asks Glaucon to imagine human beings chained inside a cave, their heads fixed so that they can look only at one wall. Shadows are cast onto that wall from an unseen world behind them. Others, outside their field of vision, carry objects before a fire, producing the shadows the prisoners take for reality. For those imprisoned viewers, the shadows are not a copy of the world but the whole of the world they know. Plato's metaphor describes a permanent human predicament: we see only what we are given to see, shadows of larger and hidden realities.

Plato's Cave, in another sense, became the optical model for later projection media. In an etching, Samuel van Hoogstraten (1627–1678) depicts a shadow play in seventeenth-century Amsterdam, where silhouettes are projected and enlarged on a wall. His viewers sit willingly where Plato's prisoners were confined by force. The image makes visible a long history of spectatorship in which people gather to watch projected appearances whose source lies elsewhere—a history that runs from shadow theater to cinema and, more obliquely, to photography itself.

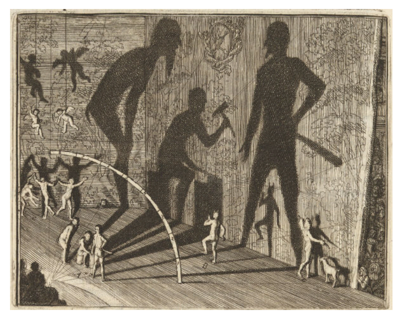


Figure 219: *Inleyding tot de Hooge Schoole der Schilderkonst*, Samuel van Hoogstraten, 1678. Hoogstraten's image of projected shadows links early modern visual entertainment to the older logic of Plato's cave: spectators see appearances cast before them while their source remains elsewhere. National Gallery of Art

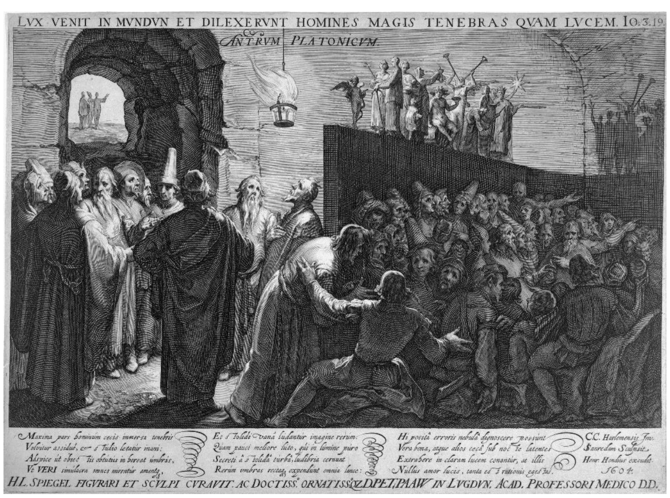


Figure 220: *Plato's Cave*, Jan Saenredam, 1604. Saenredam visualizes Plato's allegory as a drama of projection, reminding us that seeing is always mediated by the conditions under which images appear. British Museum

Modern photographers have found new ways to challenge or reinterpret Plato's Cave. The movie theater is a modern cave, where viewers willingly seat themselves to see only what the screen allows them to see.

Hiroshi Sugimoto (1948–) brought his own camera into the cinema. When the movie began, he opened the shutter of his large-format camera and exposed the film for the entire length of the film. The resulting photograph collapses moving images into a single field of light, stretching the dimension of time until the screen becomes a luminous blank. Sugimoto turns the cinema back into a meditation on projection itself: a modern cave in which time, light, and spectatorship are compressed into a still image.

Figure 221: *Paramount Theatre, Oakland*, Hiroshi Sugimoto, 1994. By leaving the shutter open for the duration of the movie, Sugimoto transforms cinema into a single photograph, making the theater itself visible while the projected film dissolves into pure light. MFA

HIROSHI SUGIMOTO is also known for seascapes that flatten the dimensions of space. In these images, the world is reduced to its most elementary structure: sea below, sky above, and the horizon between them. After seeing Sugimoto's seascapes, the horizontal bands in a Rothko can begin to look like horizons. Sugimoto thus achieves something paradoxical—painterly abstraction by photographic means, and photographic realism in a form that approaches pure abstraction.

Figure 222: *Ionian Sea, Santa Cesarea I*, Hiroshi Sugimoto, 1990. Sugimoto reduces the photographed world to sea, sky, and horizon, producing an image that is at once photographically exact and nearly abstract. MFA

ABE MORELL (1948–) turned an ordinary hotel room into Plato's Cave and, at the same time, back into a camera obscura. He blacked out the windows so that light could enter only through a pinhole in the curtain. The outside world was then projected into the room, inverted upon its interior surfaces. Morell opened the shutter of his camera and gradually recorded the faint image cast by that little light. Because so little light enters through a pinhole, the exposure had to be long. In Morell's photograph, the ancient optics of the camera obscura reappear inside modern photography itself.

Figure 223: *Camera Obscura Image of the Empire State Building in Bedroom*, Abe Morell, 1994. Morell transforms a hotel room into both Plato's cave and a camera obscura, allowing the outside world to appear upside down upon the room's interior before fixing that projection as a photograph. MFA

DIGITAL CAMERAS now allow Abe Morell to work much more quickly. An ultrasensitive, high-pixel-density camera can capture in minutes what once required hours of exposure in chemical photography. A large inkjet printer then enlarges each digital image.

DIGITAL IMAGING might seem to open the door to digital manipulation, but Morell does not walk through that door. His only manipulations lie in how he sets up the camera and the conditions under which light enters a space. He plays with the optics that cast light onto a surface, and with the surfaces that receive the image. His photographs record the natural play of light in the world, as it might be seen if the eye were simply more sensitive.

Figure 224: *View of Lower Manhattan, Sunrise*, Abe Morell, 2022. Using digital photography not to alter the image afterward but to register faint optical events more quickly, Morell preserves the camera obscura's old magic within a modern medium. National Gallery of Art

A TENT CAMERA OBSCURA took Abe Morell's camera back into the landscape. His device uses a periscope to project the surrounding world onto the ground inside a darkened tent. A digital camera then records that projected image, producing effects that are at once optical, photographic, and painterly. Morell has made such pictures in Monet's garden at Giverny, in Van Gogh's fields at Arles, and in the English landscapes of Constable. With this "camera inside a camera obscura," his photographs bring the history of the course full circle. The camera obscura shows that images can be projected before they are drawn; perspective shows that space can be organized before it is painted; photography shows that light can be fixed before it disappears. Morell combines all three histories in a single image, reconnecting the modern camera to the oldest problems of perspective, realism, and painting.

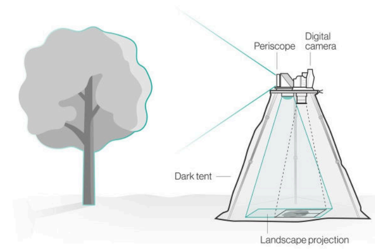


Figure 225: *Diagram of Abe Morell's tent camera obscura.* A periscope projects the surrounding landscape into the darkened tent, where a digital camera records the image cast upon the ground.

Figure 226: *Water Lilies in Monet's Garden*, Abe Morell, 2023, made with a tent camera obscura. Morell returns photography to its origins in projection while carrying it back into the landscapes of painting, turning Monet's garden into an image that is simultaneously optical event, photograph, and painterly surface.

Updated: April 15, 2026

Bibliography

- Abelardo Morell. *Camera obscura*. Bulfinch Press, New York, NY, 1st ed. edition, 2004. ISBN 0821277510
- Susan Sontag. *Susan Sontag : essays of the 1960s and 70s*. Library of America 246. Library Of America, New York, NY, 2013. ISBN 9781598532555
- Martin Kemp. *Seen/unseen : art, science, and intuition from Leonardo to the Hubble telescope*. Oxford University Press, Oxford ; New York, 2006b. ISBN 0199295727

Figure 227: *Poppy Field #s, Near Vétheuil, France* by Abe Morell, 2022, taken with tent camera obscura.

Updated: April 15, 2026